Activity 1C: Using a Button to Control a LED

Purpose: To program a button (input) to turn on a LED (output).

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, LED, button, two 330 Ω Resistor (recommened), 2 Jumper Wires.

New Components:

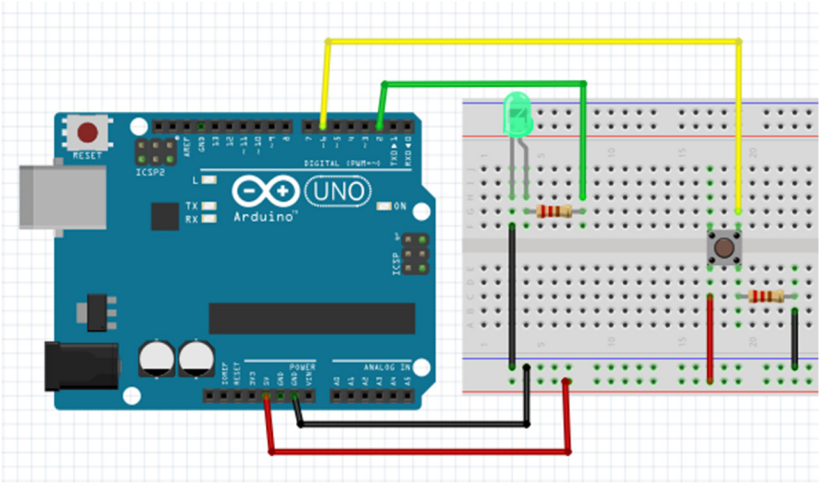
Push Button: A switch that can be used to signal a component to be turned on or off.



Resistors: Resistors resist the flow of electricity. You can use them to protect sensitive components like LEDs from too much current. The strength of a resistor (measured in ohms) is marked on the body of the resistor using small colored bands. Each color stands for a number, which you can look up using a resistor chart.

Procedure:

1. Wire the project using the following schematic.

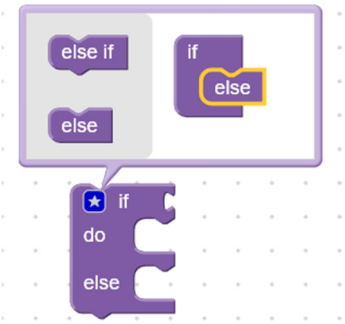


2. You can create your code in the BlocklyDuino Editor then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

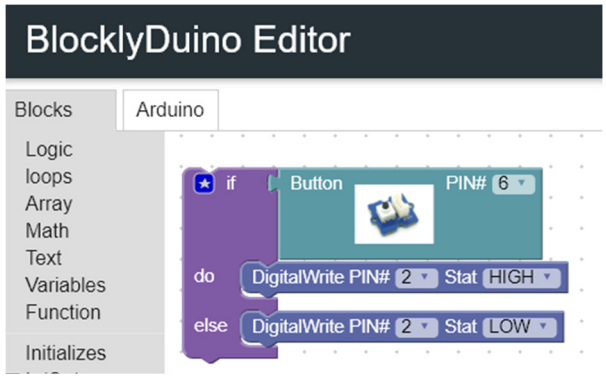
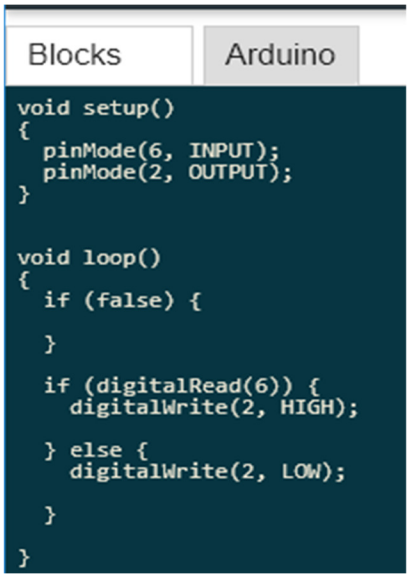
3. Once in BlocklyDuino, use the following blocks and put them into the editor to get

your button to control when the light is on and when the light is off. Watch the following video to help you with this.

- In this code we are using the **"if...else"** conditional statement. The if statement checks for a condition and executes the proceeding statement or set of statements if the condition is 'true'. We can also add an "else" to the conditions. You can add the "else" by simply selecting the blue gear on the if and dragging the else puzzle piece.



- This means it will do the first code if the condition is true or else it will do the second code.

BLOCKS	ARDUINO
	

*****SAVE your sketch to your BlocklyDuino Folder that you created*****

5. **WatchVIDEO:** [VIDEO: How to code a Button to Control a Light]

With funding from
Canada
<SaskCode>
SIEC
Saskatoon Industry Education Council

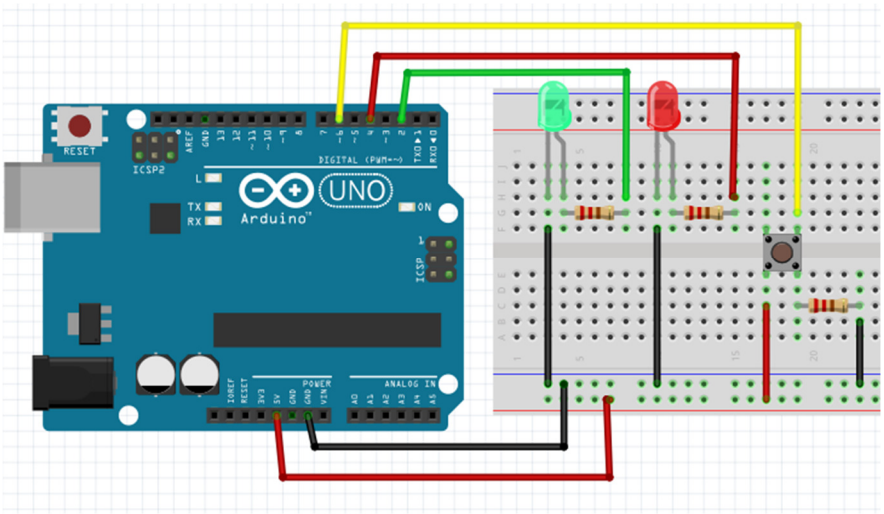
6. Copy the Arduino Code from BlocklyDuino into the Codebender website. To paste the code into the website be sure to use ctrl “v.” Hook up your arduino to your computer using the USB cable and click the upload arrow to upload the code from your board to your

6. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green “upload” button.

- Now that your code is uploaded observe the operation of the button. When the button is pressed, the light should turn on 

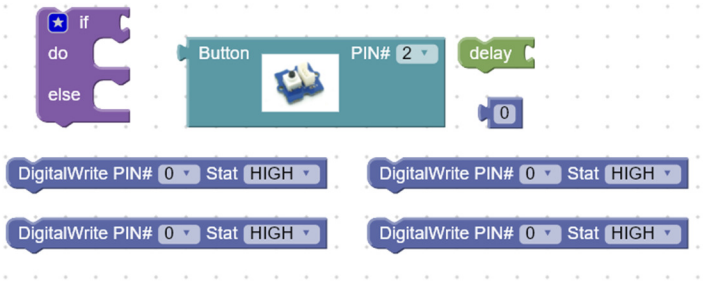
and when the button is released the light will turn off. 

7. Now you are going to add another light and a sequence to show how a light would work at a pedestrian crosswalk. When you press the button, the green light will go off for 5 seconds and the red light will go on to stop traffic so that it is safe to walk. The green light will go back on when it is no longer safe to walk.



8. Create a code to make your lights work using BlocklyDuino. You will need to use the following blocks.

With funding from
Canada
<SaskCode>
SIEC
Saskatoon Industry
Education Council



- Select the Arduino Code button to turn the blocks into code and paste the code into codebender so that you can upload to your Arduino microcontroller.

9. Once your program is loaded and your code and wiring is correct, your green light should be constantly on until you press the button. When the button is pressed, the green light will turn off and the red light will go on for 5 seconds

10. Observe your LEDs and complete your worksheet titled Assignment 1C: Using a Button to Control Lights

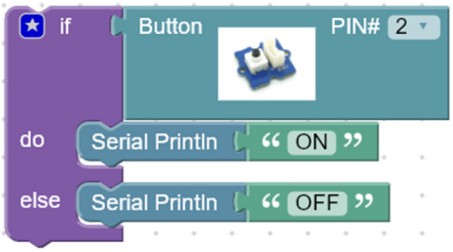
Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the leds, is the long positive arm connected by wires to the pins?
    c. Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when the your code is on?



    d. Next check the individual wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
    e. Check the button. This can be achieved by connecting it and using the serial monitor built into codebender. Load this test program:

With funding from
Canada
<SaskCode>
SIEC
Saskatoon Industry Education Council



As you hold the button down the serial monitor should change to on from off.

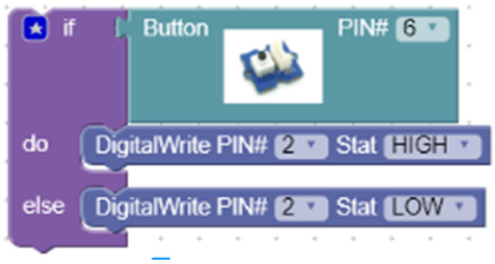
f. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

Assignment 1C: Using a Button to Control Lights

1. In the first program, when you pressed the button what happened?

2. Why did you have to hold the button down in the first program to keep the light on? What block was added to the second program that allowed you to press and release the button to keep the red light on?

3. The following is a logic block that uses a conditional statement. Explain what the conditional statement is in this program.



Extension Challenge:

4. Modify your second program using the following blocks to make the red light blink on and off ten times at intervals of 0.1 seconds when the button is pressed.

With funding from
Canada
<SaskCode>
SIEC
Saskatoon Industry Education Council



5. The green block is a loop. What is the purpose of a loop?

With funding from
Canada
<SaskCode>
SIEC
Saskatoon Industry
Education Council